

Jacob Barbulescu

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EDUCATION

Stevens Institute of Technology | Hoboken, NJ

Expected December 2026

Bachelor of Computer Science | Lawrence T. Babbio '66 Pinnacle Scholar

Relevant Coursework: Artificial Intelligence, Deep Learning, Machine Learning, Natural Language Processing, Data Structures and Algorithms, Discrete Structures, Systems Programming, Computer Human Interaction, Web Development, OOP

Awards: Reno Del Ben 1958 Endowed Scholarship, Edwin A. Stevens Scholarship, Stevens Presidential Scholarship | **GPA:** 3.9/4.0

PROJECTS & RESEARCH

Machine Learning-Driven Design of EMR Sensors | *Stevens Department of Mechanical Engineering*

June 2026 – August 2026

- Trained an XGBoost surrogate model achieving 0.949 R^2 to predict extraordinary magnetoresistance sensor sensitivity based on geometric and electrical parameters. Leveraged Bayesian Optimization to produce novel EMR designs with high sensitivities.
- Constructed a dataset of 5000+ EMR designs across 4 design categories via COMSOL simulations of EMR devices.
- Performed SHAP analysis to efficiently determine influential parameters to inform dataset expansions and design predictions.
- Produced 10 unique designs with the highest predicted sensitivities to be manufactured and physically tested.
- Designed the ML pipeline with modularity and configurability to support future expansion into additional EMR design categories.

AI-powered Postpartum Hemorrhaging Tracking Application | *HeraHealth*

September 2025 – May 2026

- Collaborated with a team of 6 to develop a medical application capable of detecting postpartum hemorrhaging in new mothers.
- Trained a DINOv3 computer vision model to detect blood volume in pictures of menstrual pads, achieving 99% accuracy.
- Developed a React-based application to enable easy cross-platform publishing, prioritizing iOS and Android mobile devices.
- Stored user photos and metrics via MongoDB with requests handled by Express. Prioritized user information security through data encryption, need-to-know access principles, and leveraging local storage on user devices.
- Wrote extensive documentation on our security framework and app design in adherence to HIPAA and ISO standards. Began FDA approval and validated our app through test cases and demo deployments in 3 US hospitals and 1 Kenyan hospital.
- Presented our work at the Stevens Innovation Expo. Gave a pitch on our B2B SAMD business model targeting hospitals.

LLM-based Mental Health Peer Support Coaching | *Stevens Department of Computer Science*

June 2025 – August 2025

- Designed PECO, an AI-powered mental health peer support coaching system where the user converses with a simulated support seeker and receives live coaching from FIDO, a custom-built feedback engine that analyzes the conversations as they occur.
- Built on Patient- Ψ , an LLM framework employing 100+ Cognitive Conceptualization Profiles to represent realistic support seekers. Fine-tuned Meta's LLaMA 4 Scout model using professional support strategies to serve as the feedback system.
- Developed custom model training scripts with Python, utilizing libraries such as HuggingFace Transformers, PyTorch, and Pandas.
- Compared various AI training methodologies, including model fine-tuning and Retrieval-Augmented Generation (RAG) Pipelines.
- Leveraged Stevens' high-power computing (HPC) clusters to expedite training for a faster, iterative design approach.
- Presented the project at the Next-Gen Healthcare Innovators Symposium 2025.

WORK EXPERIENCE

Mathnasium

Assistant Center Director

September 2025 – Present

- Create and manage student accounts, assessing progress and developing custom learning plans for over 200 active students.
- Manage the performance, satisfaction, and shifts of a team of over 30 instructors, including training and leading new employees.
- Communicate student progress and learning plans with parents through in-person chats, Zoom meetings, and email.

Lead Instructor

September 2023 – September 2025

- Conducted membership sales with prospective clients and registered new student accounts.
- Organized community events to promote community outreach, increase public awareness, and attract new clients.

SKILLS AND CERTIFICATIONS

Python Libraries: PyTorch, HuggingFace Transformers, XGBoost, scikit-learn, Pandas, Joblib, Pyplot, YAML

Machine Learning: Computer Vision, Deep Learning, Bayesian Optimization, Retrieval-Augmented Generation (RAG), SHAP

Frameworks and Tools: React, Express, Node.js, MongoDB, ImageMagick, Git, Android Studio, Three.js, TinyMCE

Languages: Python, JavaScript, Java, C++, SQL